ELHAM ROSTAMI

Doctoral Student, Université Paris-Saclay, Paris, France

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Google Scholar, **ResearchGate**, **LinkedIn**

EDUCATION

Doctoral Student (A MSCA fellow) Oct 2025 - Now

Université Paris-Saclay, QS Ranking = 70, Paris, France, in Inria-saclay / CIAMS (BOOST team)

- PhD supervisor: Prof. Taous Meriem Laleg
- Co-supervisor: Prof. Hamidou Tembine
- Research topic: Improving the training, stability, and robustness of generative AI using control theory.
- Member of Marie Skłodowska-Curie Actions (MSCA)

Master 2 in Computational Neurosciences and Neuroengineering (CNN) Sep 2024 - Jul 2025

Institut des Neurosciences Paris-Saclay - NeuroPSI CNRS, Université Paris-Saclay, QS Ranking = 70, Paris, France, following an internship in Inria-saclay (BOOST team)

- Academic advisor: prof. Sabir JACQUIR
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Selected Courses	Grade
Machine learning	19.8/20
Methods and Stimulating	17.5/20
Dynamical System	16.65/20
Closed loop Neuroscience	14.5/20

MS.c. in Biomedical Engineering-Bioelectric (Avg.Grade=18.6/20, Ranked 2nd) Oct 2018 - Sep 2021

Amirkabir University of Technology (AUT), QS Ranking = 456, Tehran, Iran

- Ranked 2nd among MS.c. students
- Thesis: *Analysis of Electroencephalogram Signal for Offline Frequency Recognition in Brain Computer Interface Speller Based on Steady State Visually Evoked Potential.*
- Supervisor: Dr. Farnaz Ghassemi
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Selected Courses	Grade
Machine Learning in Biomedicine	18.5/20
Pattern Recognition	17.4/20
Biological Signal Processing	17/20
Digital Signal Processing	20/20
Statistical Analysis of Medical Data	20/20

BS.c. in Biomedical Engineering-Bioelectric (Avg.Grade=18.19/20, Top %10) Oct 2014 - Sep 2018

Amirkabir University of Technology (AUT), QS Ranking = 456, Tehran, Iran

- Thesis: *Investigation of the Relationship between Mayers Briggs Personality Types and Internet Addiction.*
- Supervisor: Dr. Farnaz Ghassemi

High School diploma (Avg. Grade=19.91/20) Oct 2010 - Sep 2014

Baharan School, Ardabil, Iran

RESEARCH INTERESTS

Generative AI, Optimal Control, Biomedical signal processing, Biomedical image processing, Brain–Computer Interface, Brain Signal Processing.

PUBLICATIONS

Accepted Papers

- **Rostami, E.**, Laleg-Kirati, T.-M., Tembine, H. (2026). *"An Explicit Surrogate for Gaussian Mixture Flow Matching with Wasserstein Gap Bounds"*, 65th IEEE Conference on Decision and Control (CDC) **accepted** arXiv
- **Rostami, E.**, Einizade, A., Laleg-Kirati, T.-M. (2026). *"SpaTeoGL: Spatiotemporal Graph Learning for Interpretable Seizure Onset Zone Analysis from Intracranial EEG"*, 34th European Signal Processing Conference (EUSIPCO) arXiv
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2023). *"Transfer Learning assisted PodNet for Stimulation Frequency Detection in Steady state visually evoked potential-based BCI Spellers"*, Brain-Computer Interfaces, 10, 38-49. (IF: 2.6) doi
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2022). *"Canonical Correlation Analysis of Task Related Components as a noise-resistant method in Brain-Computer Interface Speller Systems based on Steady-State Visual Evoked Potential"*, Biomedical Signal Processing and Control, 73, 103449. (IF: 5.7) doi
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2022). *"Improving the classification of real-world SSVEP data in brain-computer interface speller systems using deep convolutional neural networks"*, Frontiers in Biomedical Technologies, 9(4), 248-254. (IF: 0.5) doi PDF
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2022). *"Improving the classification of Real-World SSVEP data in Brain-Computer Interface Speller systems using Deep Convolutional Neural Networks"*, In 5th Iranian Symposium on Brain Mapping Updates (ISBM), 8, 19-21. doi PDF
- **Rostami, E.**, Ghassemi, F., (2020). *"Investigation of the effect of Attention in BCI systems based on SSVEP"*, In 8th Basic and Clinical Neuroscience Congress, Iran University of Medical Sciences. doi PDF

Papers Under review

- **Rostami, E.**, Laleg-Kirati, T.-M., Tembine, H. (2026). *"From Local Gaussian Transport to Global Mixture Flow Matching: Explicit Surrogates, Mixture-Level Bounds, and Empirical Validation"*

RESEARCH/JOB EXPERIENCE

- Internship in Inria-saclay, under the supervision of Prof. Taous-Meriem Laleg Kirati, **Project-team:** Bio-informed mOnitoring and Optimization for enhanced Sport and healTh (BOOST). Title: "Seizure Onset Zone Localization Using Visibility Graphs-Based Features." Jan 2025 - Jul 2025
- Senior Technical Mentor in AI: Python programming, Machine Learning, Computer Vision, AI Algorithms, *AIOLearn Academy*, Tehran, Iran. Feb 2023 - Jul 2023
- Judge and Deputy of Programming and Artificial Intelligence Competitions in the National Competitions of Creativity, Innovation, and Entrepreneurship, *AIOLearn Academy*, Tehran, Iran. Jun 2023 - Jul 2023
- Project Manager and Senior Programmer in "Personnel Identification System with Time Entry/Time Out", *AIOLearn Academy*, Tehran, Iran. May 2023 - Jun 2023
- Researcher at U: The Mind Company (development of a Message Passing Neural Network (MPNN) for Uncertainty Quantification), Columbus, Ohio. Jan 2022 - May 2022
- Internship: EEG signal recording and processing, National Brain Mapping Laboratory (NBML), Tehran, Iran. Jan 2022 - Feb 2022
- Research Assistant (RA): Brain Signal Processing, Revising Papers and Proposals, *Amirkabir University of Technology (AUT)*, Tehran, Iran. Dec 2021 - Jun 2022
- Teacher Assistant (TA): Research Methodology and Technical Writing, Computational Neuroscience Lab, *Amirkabir University of Technology (AUT)*, Tehran, Iran. Sep 2020 - Fab 2021

- Teacher Assistant (TA): Bioelectrical Phenomena, Action-potential Generation in excitable tissues and Electric Field Effects on the Body, Computational Neuroscience Lab, *Amirkabir University of Technology (AUT)*, Tehran, Iran. Sep 2019 - Jan 2020

PROJECTS (For more information let me know, I will share the details with you)

Academic Projects (Master and Bachelor)

- EEG signal quality analysis by Beacon Biosignals (The Dreem Headband), Score: 75.406% .
- Study on "Binocular Rivalry", Dynamical systems & Neuroscience, Universit´e Paris-Saclay.
- EEG signal Preprocessing, Feature extraction, and Classifications techniques
  "specifically SSVEP signals, using the EEGLAB toolbox and EEG signal recording and processing"
- Extraction of stimulation-induced Evoked Potentials from EEG signal by synchronous averaging
- Medical Engineering Internship at Mostafa Khomeini Hospital in Tehran (Operating Room Department)
  "An internship in summer semester with forming valuable connections with doctors and nurses."

INVITED REVIEWER

Conferences: IEEE Conference on Decision and Control (CDC) and The European Signal Processing Conference (EUSIPCO).

LICENSES AND CERTIFICATIONS

- **Certificate of Attendance and Oral Presentation** — Presented the paper "SpaTeoGL: Spatiotemporal Graph Learning for Interpretable Seizure Onset Zone Analysis from Intracranial EEG" **at EUSIPCO 2026**, Bruges, Belgium. Aug 31 - Sep 4
- National Competition of Innovation and Invention in Programming and AI, *AIOLearn Academy*, Tehran, Iran. Mar 2023
- Internship: EEG signal recording and processing, NBML, Iran. Jan 2022
- Applied Artificial Intelligence (based on Python), Tehran University, Iran. Jan 2019
- Brain Signal Processing in EEGLAB toolbox, Amirkabir University of Technology (AUT). Jan 2018

COMPUTER SKILLS

Programming Languages
Advanced: Python, MATLAB.
Intermediate: HTML, CSS, PHP, SQL.

Relevant Libraries
Keras, Tensorflow, OpenCV, scikit-learn, EEGLAB toolbox, MATLAB GUI.

Other Software and Tools
Photoshop, LaTeX.

LANGUAGES

English: Professional Sufficiency.
Farsi: Native Language.
Turkish: Native Language.
French: A1 level, currently in progress.

AWARDS AND HONORS

- Distinguished student in master's degree, Biomedical Engineering Department, Amirkabir University of Technology, 2021.

- Ranked 2nd among MS.c. students in Biomedical Engineering Department, AUT, 2021
- Top %10 in BS.c. and Direct Admission to graduate school (M.Sc.) of Biomedical Engineering Department, AUT, without taking the national entrance exam, as a reward for academic records and achievements, 2018.
- Ranked Top %2 in the National University Entrance Exam among +222000 students, 2014.

REFERENCES

Associate Professor Taous-Meriem Laleg Kirati (Supervisor)
Electrical and Computer Engineering
INRIA Researcher, INRIA Paris-Saclay
Email: *taous-meriem.laleg@inria.fr*

Associate Professor Hamidou Tembine (Supervisor)
Department of Electrical Engineering and Computer Science
University of Quebec in Trois-Rivieres (UQTR), Quebec, Canada
Email: *hamidou.tembine@uqtr.ca*

Associate Professor Farnaz Ghassemi (previous Supervisor)
Department of Biomedical Engineering (Bioelectric)
Amirkabir University of Technology
Email: *ghassemi@aut.ac.ir*

Dr. Zahra Tabanfar Post Doc (Consultant, the percentage of cooperation $\geq 90\%$)
Technische Universität Dresden
Kognitive Neurophysiologie am Universitätsklinikum Carl Gustav Carus
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